

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only

Budecort Respules 0.5mg/2ml Respirator Suspension

COMPOSITION

Each 2 ml respule contains

Budesonide BP0.5 mg

DOSAGE FORM

Solution for inhalation via a nebulizer.

PRODUCT DESCRIPTION

A white to off white homogenous redispersible suspension filled in 2 ml FFS respules.

On visual examination, there is no sign of leakage or physical damage.

PHARMACOLOGY

Pharmacodynamics

Budesonide is a glucocorticosteroid which possesses a high local anti-inflammatory action, with a lower incidence and severity of adverse effects than those seen with oral corticosteroids.

Topical anti-inflammatory effect

The exact mechanism of action of glucocorticosteroids in the treatment of asthma is not fully understood. Anti-inflammatory actions, such as inhibition of inflammatory mediator release and inhibition of cytokine-mediated immune response are probably important.

A clinical study in asthmatics comparing inhaled and oral budesonide at doses calculated to achieve similar systemic bioavailability demonstrated statistically significant evidence of efficacy with inhaled but not oral budesonide compared with placebo. Thus, the therapeutic effect of conventional doses of inhaled budesonide may be largely explained by its direct action on the respiratory tract.

In a provocation study pre-treatment with budesonide for four weeks has shown decreased bronchial constriction in immediate as well as late asthmatic reactions.

Onset of effect

After a single dose of orally inhaled budesonide, delivered via dry powder inhaler, improvement of the lung function is achieved within a few hours. After therapeutic use of orally inhaled budesonide delivered via dry powder inhaler, improvement in lung function has been shown to occur within 2 days of initiation of treatment although maximum benefit may not be achieved for up to 4 weeks.

Airway reactivity

Budesonide has also been shown to decrease airway reactivity to histamine and methacholine in hyperreactive patients.

Exercise-induced asthma

Therapy with inhaled budesonide has effectively been used for prevention of exercise-induced asthma.

Growth

Limited data from long-term studies suggest that most children and adolescents treated with inhaled budesonide ultimately achieve their adult target height. However, an initial small but transient reduction in growth (approximately 1 cm) has been observed. This generally occurs within the first year of treatment (see section Warning and precaution).

Pharmacokinetics

Budesonide undergoes an extensive biotransformation in the liver, to metabolites of low glucocorticosteroid activity. The glucocorticosteroid activity of the major metabolites, 6 β -hydroxybudesonide and 16 α -hydroxyprednisolone, is less than 1% of that of budesonide. The metabolism of budesonide is primarily mediated by CYP3A4, one of the cytochrome p450 enzymes.

In a study, 100 mg ketoconazole taken twice daily, increased plasma levels of concomitantly administered oral budesonide (single dose of 10 mg) on average, by 7.8-fold. Information about this interaction is lacking for inhaled budesonide, but marked increases in plasma levels could be expected.

Of the fraction of budesonide which is swallowed, approximately 90% is inactivated at first passage through the liver. The maximal plasma concentration after inhalation of 1 mg budesonide, delivered via dry powder inhaler, is about 3.5 nmol/L and is reached after about 20 minutes.

INDICATIONS

Budecort Respules contain the potent non-halogenated corticosteroid, budesonide for use in bronchial asthma in patients where use of a pressurized inhaler or dry powder formulation is unsatisfactory or inappropriate.

ROUTE OF ADMINISTRATION

Solution for oral inhalation

DOSAGE AND ADMINISTRATION

Budecort respules should be administered from a suitable nebuliser. The dose delivered to the patient varies between 40-60% of the nominal dose depending on the nebulising equipment used. The nebulisation time and the dose delivered are dependent on flow rate, volume of nebuliser chamber and volume fill. A suitable fill for most nebulisers is 2-4mL. Some sedimentation may occur during storage of Budecort respules. If this does not readily resuspend completely upon shaking, the RESPULE should be discarded. Dosage initially, or during periods of severe asthma, or while reducing oral corticosteroids Adults 1-2mg twice daily. Children 0.5-1mg twice daily.

Maintenance The maintenance dose should be individualised and should be the lowest dose, which keeps the patient symptom-free. Recommended doses are: Adults 0.5-1mg twice daily. Children 0.25-0.5mg twice daily.

Patient Instructions It is essential that the patient be instructed that Budecort respules is a preventative agent, which must be taken regularly and is not to be used as sole therapy to relieve an acute asthma attack. The patient should be instructed in the proper use of the inhaler device considered appropriate for his/her particular needs. A full set of instructions are provided with each pack of Budecort respules. Patients also receiving bronchodilators by inhalation should be advised to use the bronchodilator before Budecort

respules in order to enhance its penetration into the bronchial tree. Several minutes should elapse between the use of the two inhalers.

CONTRAINDICATIONS

History of hypersensitivity to budesonide or any of the excipients.

WARNINGS AND PRECAUTIONS

Special care is needed in patients with pulmonary tuberculosis and viral infections of the airways.

Non steroid-dependent patients:

A therapeutic effect is usually reached within 10 days. In patients with excessive mucus secretion in the bronchi, a short (about 2 weeks) additional oral corticosteroid regimen can be given initially. After the course of the oral drug, budesonide respules alone should be sufficient therapy.

Steroid-dependent patients:

When transfer from oral corticosteroid to treatment with budesonide respules is initiated, the patient should be in a relatively stable phase. Budesonide respules is then given, in combination with the previously used oral steroid dose, for about 10 days.

After that, the oral steroid dose should be gradually reduced (by, for example, 2.5 mg prednisolone or the equivalent each month), to the lowest possible level. In many cases, it is possible to completely substitute budesonide respules for the oral corticosteroid.

During transfer from oral therapy to budesonide respules, a generally lower systemic corticosteroid action will be experienced, which may result in the appearance of allergic or arthritic symptoms such as rhinitis, eczema and muscle and joint pain. Specific treatment should be initiated for these conditions. A general insufficient glucocorticosteroid effect should be suspected if, in rare cases, symptoms such as tiredness, headache, nausea and vomiting should occur. In these cases a temporary increase in the dose of oral glucocorticosteroids is sometimes necessary.

As with other inhalation therapy, paradoxical bronchospasm may occur, with an immediate increase in wheezing after dosing. If a severe reaction occurs, treatment should be reassessed and an alternative therapy instituted if necessary.

Prolonged treatment with high doses of inhaled corticosteroids, particularly higher than the recommended doses, may result in clinically significant adrenal suppression. Additional systemic corticosteroid cover should be considered during periods of stress or elective surgery.

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged periods. These effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract and glaucoma.

It is important, therefore, that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control of asthma is maintained.

It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is slowed, therapy should be reviewed, with the aim of reducing the dose of inhaled corticosteroid, if possible, to the lowest dose at which effective control of asthma is maintained. In addition, consideration should be given to referring the patient to a paediatric respiratory specialist.

Budesonide respules is not intended for rapid relief of acute episodes of asthma where an inhaled short-acting bronchodilator is required. If patients find short-acting bronchodilator treatment ineffective, or they need more inhalations than usual, medical attention must be sought. In this situation consideration should be given to the need for or an increase in their regular therapy, e.g., higher doses of inhaled budesonide or the addition of a long-acting beta agonist, or for a course of oral glucocorticosteroid.

Reduced liver function may affect the elimination of glucocorticosteroids. The plasma clearance following an intravenous dose of budesonide however was similar in cirrhotic patients and in healthy subjects. After oral ingestion systemic availability of budesonide was increased by compromised liver function due to decreased first pass metabolism. The clinical relevance of this to treatment with budesonide respules is unknown as no data exist for inhaled budesonide, but increases in plasma levels and hence an increased risk of systemic adverse effects could be expected.

Concomitant treatment with ketoconazole and itraconazole or other potent CYP3A4 inhibitors should be avoided. If this is not possible, the time interval between administration of the interacting drugs should be as long as possible. A reduction in the dose of budesonide should also be considered.

The nebuliser chamber should be cleaned after every administration. Wash the nebuliser chamber and mouthpiece or face-mask in hot water using a mild detergent. Rinse well and dry, by connecting the nebuliser chamber to the compressor or air inlet.

DRUG INTERACTIONS

The metabolism of budesonide is primarily mediated by CYP3A4, one of the cytochrome p450 enzymes. Inhibitors of this enzyme, e.g. ketoconazole and itraconazole, can therefore increase systemic exposure to budesonide, (see Section Warnings and Special Precautions and Section Pharmacokinetic Properties). Other potent inhibitors of CYP3A4 are also likely to markedly increase plasma levels of budesonide.

PREGNANCY AND LACTATION

Data on approximately 2000 exposed pregnancies indicate no increased teratogenic risk associated with the use of inhaled budesonide. In animal studies, glucocorticosteroids have been shown to induce malformations. This is not likely to be relevant for humans given recommended doses, but therapy with inhaled budesonide should be regularly reviewed and maintained at the lowest effective dose.

The administration of budesonide during pregnancy requires that the benefits for the mother be weighed against the risk for the foetus. Inhaled glucocorticosteroids should be considered in preference to oral glucocorticosteroids because of the lower systemic effects at the doses required to achieve similar pulmonary responses. There is no information regarding the passage of budesonide into breast milk.

UNDESIRABLE EFFECTS

Clinical trials, literature reports and post-marketing experience suggest that the following adverse drug reactions may occur:

Common (>1/100, <1/10)	• Mild irritation in the throat • Candida infection in the oropharynx • Hoarseness • Coughing
Rare (>1/10 000, <1/1 000)	• Nervousness, restlessness, depression, behavioural disturbances • Immediate and delayed hypersensitivity reactions including rash, contact dermatitis, urticaria, angioedema and bronchospasm • Skin bruising

The candida infection in the oropharynx is due to drug deposition. Advising the patient to rinse the mouth out with water after each dosing will minimise the risk.

As with other inhalation therapy, paradoxical bronchospasm may occur in very rare cases (see Section Warning and Precaution).

Systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged periods. These effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract and glaucoma. The effect is probably dependent on dose, exposure time, concomitant and previous steroid exposure, and individual sensitivity.

Facial skin irritation has occurred in some cases when a nebuliser with a face mask has been used. To prevent irritation, the facial skin should be washed with water after use of the face mask.

OVERDOSAGE

Budecort Respules contains 0.1 mg/ml disodium edetate which has been shown to cause bronchoconstriction at levels above 1.2 mg/ml. Acute overdose with Budecort should not present a clinical problem.

STORAGE

Store below 30°C.

Protect from light.

Do not freeze.

SHELF LIFE

2 Years

PRESENTATION

Budecort Respules 0.5 mg

Cartons containing 4 triple laminated pouches each having 5 FFS respules of 2.0 ml.

REGISTRATION HOLDER IN MALAYSIA

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